

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Alternative: Attempt at using R values (1) $R_{\text{concrete}} = \frac{0.12}{0.7} = 0.17$ $R_{\text{underlay}} = \frac{0.01}{0.035} = 0.29$ $R_{\text{carpet}} = \frac{0.01}{0.045} = 0.22$ Total R value = $0.68 \text{ [K m}^2 \text{ W}^{-1}]$ (1) So $R_T = \frac{\frac{\Delta T}{\phi}}{A} \rightarrow \phi = \frac{\Delta T A}{R_T}$ $\phi = \frac{14 \times 60}{0.68} = 1235 \text{ [W]}$ so claim is reasonable (1)						
			Question 10 total	6	9	5	20	10	0